WEEK 1 · LESSON 6 OF 24 · HANDOUT

Week 1 Project: A Shop Sales Calculator

Study time: **about 60 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=UmNKiV7Lekk — open the lesson online to watch inside the course.

Your goal in this lesson: Combine variables, lists, dictionaries, loops, decisions, functions and files into one small program that solves a real problem.

The task

A small shop in Kigali records each sale by hand. The owner wants a simple program that:

1. calculates each sale's total with 18% VAT,
2. gives a 5% discount on sales of 50,000 RWF or more,
3. prints a daily report (total, number of sales, best-selling product),
4. saves the sales to a CSV file.

You will build it step by step. Write each step in its own Colab cell and run it before moving on.

Step 1 — The data

```
prices = {"Rice 5kg": 7500, "Cooking oil 1L": 3200, "Sugar 1kg": 1600,
          "Milk 1L": 800, "Airtime 1000": 1000}

sales = [
    {"product": "Rice 5kg", "quantity": 4, "payment": "MoMo"},
    {"product": "Milk 1L", "quantity": 6, "payment": "Cash"},
    {"product": "Cooking oil 1L", "quantity": 2, "payment": "MoMo"},
    {"product": "Rice 5kg", "quantity": 7, "payment": "Card"},
    {"product": "Sugar 1kg", "quantity": 3, "payment": "Cash"},
    {"product": "Airtime 1000", "quantity": 5, "payment": "MoMo"},
]
```

Step 2 — Functions for one sale

```
VAT = 0.18

def sale_total(product, quantity):
    # Price x quantity, plus VAT, minus 5% if the sale is 50,000 RWF or more.
    subtotal = prices[product] * quantity
    total = subtotal * (1 + VAT)
    if total >= 50000:
        total = total * 0.95
    return round(total)

print(sale_total("Rice 5kg", 4))   # 35400
print(sale_total("Rice 5kg", 7))   # 58852 (discount applied)
```

Step 3 — The daily report

```
def daily_report(sales):
    grand_total = 0
    quantity_by_product = {}
    for s in sales:
        total = sale_total(s["product"], s["quantity"])
        s["total"] = total
        grand_total += total
        quantity_by_product[s["product"]] = quantity_by_product.get(s["product"], 0) +
s["quantity"]

    best = max(quantity_by_product, key=quantity_by_product.get)
    momo = sum(1 for s in sales if s["payment"] == "MoMo")

    print("=== DAILY REPORT ===")
    print(f"Sales recorded : {len(sales)}")
    print(f"Total income   : {grand_total:,} RWF")
    print(f"Best seller    : {best} ({quantity_by_product[best]} units)")
    print(f"Paid by MoMo   : {momo} of {len(sales)}")
    return grand_total

daily_report(sales)
```

New tools here: - `dict.get(key, 0)` returns 0 when the key is not there yet — perfect for counting. - `max(d, key=d.get)` finds the key with the largest value. - `sum(1 for s in sales if ...)` counts items that meet a condition.

Step 4 — Save to CSV

```
import csv

with open("sales_today.csv", "w", newline="") as f:
    writer = csv.DictWriter(f, fieldnames=["product", "quantity", "payment", "total"])
    writer.writeheader()
    writer.writerows(sales)

print(open("sales_today.csv").read())
```

Step 5 — Make it your own (required)

Add **at least two** of these improvements:

- A function `add_sale(product, quantity, payment)` that refuses unknown products with a clear message.
- A report line showing income **by payment method**.
- A loyalty rule: MoMo payments above 20,000 RWF get an extra 1% off.
- Read the sales from a CSV file instead of typing them in the code.

Checklist

- Every function has a clear name and a one-line comment explaining it.
- You tested each function with at least two inputs, including an edge case (for example, exactly 50,000 RWF).
- The report prints neatly with thousands separators.
- The CSV file opens correctly (download it from the Colab folder and open it in Excel or Google Sheets).

Remember this: Real programs are small pieces — data, functions, loops and decisions — joined together. Build one step at a time and test each step before moving on.

Name: _____ Date: _____

Activity

Task 1

Complete Steps 1–5 in one notebook. Then share the notebook (Colab → Share → "Anyone with the link can view") with a classmate and ask them to find one bug or one way to make the report clearer.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What does `quantity_by_product.get(product, 0)` return when the product is not in the dictionary yet?
   A. The product name
   B. 0
   C. An error
   D. None
2. A sale's total with VAT is exactly 50,000 RWF. With the rule `if total >= 50000`, is the discount applied?
   A. Only on Mondays
   B. No
   C. It causes an error
   D. Yes, because >= includes 50,000
3. What does `max(d, key=d.get)` return?
   A. The largest value
   B. The number of keys
   C. The first key
   D. The key whose value is largest
4. Why build the program step by step in separate cells?
   A. So you can test each part before adding the next
   B. To hide errors
   C. To make it longer
   D. Colab requires it
5. What does `csv.DictWriter` write?
   A. A PDF
   B. Dictionaries as rows, using the fieldnames as columns
   C. Pictures
   D. Only numbers
6. Which is a good test case for the discount rule?
   A. Only one random sale
   B. Only very small sales
   C. A sale exactly at the 50,000 RWF limit, plus one below and one above
   D. No tests are needed

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